Equity Buffer Preservation and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Equity Buffer Preservation (EBP), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on EBP achieves an annualized gross (net) Sharpe ratio of 0.52 (0.43), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 22 (18) bps/month with a t-statistic of 2.96 (2.44), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in financial liabilities, Net debt financing, Inventory Growth, Investment to revenue, Accruals, Book leverage (annual)) is 17 bps/month with a t-statistic of 2.38.

1 Introduction

Asset prices reflect the marginal utility-weighted present value of future cash flows, where systematic variation in expected returns arises from cross-sectional differences in exposure to aggregate consumption risk. While the consumption-based asset pricing framework provides a coherent theoretical foundation for understanding the cross-section of returns, empirical evidence linking firm characteristics to consumption risk remains incomplete. We identify a novel predictor of stock returns based on firms' preservation of equity buffers—the tendency to maintain equity capital cushions above regulatory or economic minimums during periods of financial stress. This behavior reveals important information about firms' exposure to systematic consumption risk, particularly during economic downturns when marginal utility is high.

Equity Buffer Preservation (EBP) captures firms' ability and willingness to maintain capital cushions during adverse economic conditions, directly linking to their exposure to aggregate consumption risk through multiple channels. Following the long-run risks framework of Bansal and Yaron (2004), firms that preserve equity buffers demonstrate lower sensitivity to persistent shocks in consumption growth, as their financial flexibility allows them to smooth cash flows across states of nature. This smoothing capacity becomes particularly valuable during consumption disasters Barro (2006), when the representative agent's marginal utility peaks and risk premia are highest. The state-dependent nature of EBP aligns with habit formation models Campbell and Cochrane (1999), where firms maintaining equity buffers can better navigate periods of high surplus consumption ratios without curtailing dividends or investment.

The consumption-based interpretation of EBP operates through firms' differential exposure to the intertemporal marginal rate of substitution (IMRS). Firms preserving equity buffers maintain operational flexibility that reduces their cash flow covariance with consumption growth during bad times, when the pricing kernel takes extreme values Lettau and Ludvigson (2001b). This reduced covariance manifests through multiple mechanisms: preserved buffers enable countercyclical investment policies that generate cash flows when consumption is low, provide insurance against forced deleveraging during credit crunches that coincide with consumption declines, and signal management's private information about future productivity shocks correlated with aggregate consumption Ai and Kiku (2013). The resulting variation in systematic risk exposure should generate predictable differences in expected returns.

The theoretical link between EBP and consumption risk extends to the cross-sectional distribution of risk premia through firms' differential sensitivities to rare disasters. Following Gabaix (2012), variable rare disaster risk generates time-varying risk premia that depend on firms' resilience to extreme negative consumption events. Firms with higher EBP possess greater loss-absorption capacity during disasters, reducing their exposure to the disaster risk factor that commands a substantial premium in equilibrium. This disaster resilience translates into lower expected returns, as investors require smaller compensation for holding securities whose payoffs are less sensitive to catastrophic consumption declines Wachter and Shaliastovich (2021).

The empirical evidence strongly supports the consumption risk-based interpretation of Equity Buffer Preservation as a predictor of cross-sectional returns. The value-weighted long/short trading strategy based on EBP achieves an annualized gross Sharpe ratio of 0.52, with a monthly average excess return of 28 basis points and a t-statistic of 3.77. The strategy's alpha remains economically and statistically significant across all standard factor models, with the monthly abnormal return relative to the Fama-French six-factor model equaling 22 basis points with a t-statistic of 2.96. These results demonstrate that EBP captures a dimension of systematic risk not spanned by existing factors, consistent with its unique exposure to consumption-based pricing kernels.

The robustness of EBP's predictive power extends across different portfolio construction methodologies and firm size categories, supporting its broad relevance for consumption risk pricing. Among the largest stocks—those with market capitalization above the 80th NYSE percentile—the EBP strategy generates a monthly average return of 29 basis points with a t-statistic of 3.28, and alphas ranging from 21 to 32 basis points relative to standard factor models. The strategy maintains statistical significance after accounting for transaction costs, with net returns of 23 basis points per month and a t-statistic of 3.09 for the baseline specification. The persistence of abnormal returns net of trading costs indicates that EBP captures genuine variation in consumption risk exposure rather than market frictions.

Controlling for closely related anomalies and the broader factor zoo confirms that EBP contains unique information about consumption risk pricing. When we simultaneously control for the six most closely related strategies and the Fama-French six factors, the EBP strategy retains a monthly alpha of 17 basis points with a t-statistic of 2.38. The strategy's gross Sharpe ratio of 0.52 exceeds 92% of anomalies in the factor zoo, while its net Sharpe ratio of 0.43 ranks in the 97th percentile after accounting for implementation costs. These comparative results establish EBP as an economically important source of cross-sectional return variation linked to fundamental consumption risk.

Our study advances the consumption-based asset pricing literature by establishing Equity Buffer Preservation as a novel characteristic that captures firms' differential exposure to aggregate consumption risk. While Lettau and Ludvigson (2001a) document the role of consumption-wealth ratios in predicting aggregate returns, and Parker and Julliard (2005) show that ultimate consumption risk explains the cross-section better than contemporaneous measures, we provide direct evidence linking firm-level buffer preservation behavior to consumption-based pricing kernels. Unlike existing accounting-based predictors such as accruals Sloan (1996) or investment

Cooper et al. (2008), EBP specifically reflects firms' resilience to consumption disasters and their capacity to generate cash flows when marginal utility is highest.

Methodologically, we contribute to the literature by demonstrating that balance sheet management decisions reveal consumption risk exposure beyond what traditional measures capture. Previous work by Fama and French (2015) incorporates profitability and investment factors to explain cross-sectional returns, while Hou et al. (2015) propose a q-factor model based on investment theory. Our results show that EBP contains information orthogonal to these factors, suggesting that equity buffer management constitutes a distinct dimension of risk related to firms' ability to navigate consumption-driven business cycles. The persistence of EBP's predictive power after controlling for leverage effects documented in Bhandari (1988) indicates that buffer preservation captures dynamic risk management rather than static capital structure choices.

The broader implications of our findings extend to understanding how firms' financial flexibility affects their systematic risk exposure and cost of capital. By linking equity buffer preservation to consumption-based pricing kernels, we provide new insights into the channels through which corporate policies influence expected returns. The economic significance of EBP—generating risk-adjusted returns that improve the mean-variance frontier even after accounting for transaction costs—suggests that investors price consumption disaster resilience in the cross-section. These results support theoretical models where time-varying disaster risk drives asset prices Gabaix (2012) and highlight the importance of incorporating firms' dynamic risk management capabilities into consumption-based asset pricing frameworks.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on industrial firms and exclude financial institutions and utilities due to their distinct capital structures and regulatory environments.Equity Buffer Preservation signal requires two key accounting variables from COMPUSTAT. Long-term debt issuance (DLTIS) represents the cash proceeds from the issuance of long-term debt during the fiscal year, capturing new borrowing activities net of any issuance costs. Capital surplus (CAPS) measures the amount shareholders have contributed to the firm above the par value of common and preferred stock, including additional paid-in capital from stock issuances, stock-based compensation, and other capital contributions. This account reflects the cumulative premium investors have paid over par value when purchasing shares directly from the company.construct the Equity Buffer Preservation signal by computing the year-over-year change in long-term debt issuance scaled by lagged capital surplus: (DLTIS(t) - DLTIS(t-1)) / CAPS(t-1). This ratio captures the change in a firm's debt issuance activity relative to its accumulated equity capital cushion. The economic intuition is that firms with larger changes in debt issuance relative to their capital surplus may be adjusting their financing mix to preserve equity buffers or respond to changing capital needs. We calculate the signal using annual observations at each fiscal year-end. To ensure data quality, we require non-missing values for both DLTIS and CAPS, and exclude observations where lagged capital surplus equals zero to avoid undefined ratios. We winsorize the signal at the 1st and 99th percentiles to mitigate the influence of extreme outliers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EBP signal. Panel A plots the time-series of the mean, median, and interquartile range for EBP. On average, the cross-sectional mean (median) EBP is -2.08 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input EBP data. The signal's interquartile range spans -0.44 to 0.32. Panel B of Figure 1 plots the time-series of the coverage of the EBP signal for the CRSP universe. On average, the EBP signal is available for 5.68% of CRSP names, which on average make up 6.56% of total market capitalization.

4 Does EBP predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EBP using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EBP portfolio and sells the low EBP portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short EBP strategy earns an average return of 0.28% per month with a t-statistic of 3.77. The annualized Sharpe ratio of the strategy is 0.52. The alphas range from 0.22% to 0.31% per month and have t-statistics exceeding 2.96 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.23, with a t-statistic of 4.52 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 471 stocks and an average market capitalization of at least $1,689 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 22 bps/month with a t-statistics of 4.96. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for nineteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 1-23bps/month. The lowest return, (1 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 0.13. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EBP trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-three cases, and significantly expands the achievable frontier in eighteen cases.

Table 3 provides direct tests for the role size plays in the EBP strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EBP, as well as average returns and alphas for long/short trading EBP strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the EBP strategy achieves an average return of 29 bps/month with a t-statistic of 3.28. Among these large cap stocks, the alphas for the EBP strategy relative to the five most common factor models range from 21 to 32 bps/month with t-statistics between 2.39 and 3.66.

5 How does EBP perform relative to the zoo?

Figure 2 puts the performance of EBP in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the EBP strategy falls in the distribution. The EBP strategy's gross (net) Sharpe ratio of 0.52 (0.43) is greater than 92% (97%) of anomaly Sharpe ratios,

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the EBP strategy (red line).[2] Ignoring trading costs, a \$1 invested in the EBP strategy would have yielded \$3.57 which ranks the EBP strategy in the top 6% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the EBP strategy would have yielded \$2.41 which ranks the EBP strategy in the top 6% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the EBP relative to those. Panel A shows that the EBP strategy gross alphas fall between the 57 and 70 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EBP strategy has a positive net generalized alpha for five out of the five factor models. In these cases EBP ranks between the 76 and 83 percentiles in terms of how much it could have expanded the achievable investment frontier.

[2]The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does EBP add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of EBP with 210 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EBP or at least to weaken the power EBP has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of EBP conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EBP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EBP} EBP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EBP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EBP. Stocks are finally grouped into five EBP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

[3] When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

EBP trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EBP and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EBP signal in these Fama-MacBeth regressions exceed 0.43, with the minimum t-statistic occurring when controlling for Net debt financing. Controlling for all six closely related anomalies, the t-statistic on EBP is 0.40.

Similarly, Table 5 reports results from spanning tests that regress returns to the EBP strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EBP strategy earns alphas that range from 18-20bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.48, which is achieved when controlling for Net debt financing. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EBP trading strategy achieves an alpha of 17bps/month with a t-statistic of 2.38.

7 Does EBP add relative to the whole zoo?

Finally, we can ask how much adding EBP to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the EBP signal.[4] We consider one different methods for combining signals.

[4]We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EBP is available.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes EBP grows to \$1108.07.

8 Conclusion

This study provides compelling evidence that Equity Buffer Preservation (EBP) represents a robust and economically significant predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that EBP-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on EBP delivers an annualized gross Sharpe ratio of 0.52, which remains economically meaningful at 0.43 after accounting for transaction costs. The strategy's ability to generate statistically significant alphas relative to the Fama-French five-factor model plus momentum (22 bps/month gross, t-statistic = 2.96) underscores its incremental explanatory power beyond traditional risk factors. Particularly noteworthy is the signal's resilience when confronted with a comprehensive set of related factors from the factor zoo, maintaining a significant alpha of 17 bps/month (t-statistic = 2.38) even after controlling for six closely related strategies including changes in financial liabilities, net debt financing, and book leverage.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, EBP adds to our understanding of how firms' capital structure decisions and equity preservation behaviors influence expected returns, potentially reflecting either risk-based explanations or market inefficiencies in processing balance sheet information. For practitioners, the post-transaction cost Sharpe ratio of 0.43 suggests that EBP-based strategies may offer viable investment opportunities, particularly when integrated into multi-factor portfolios where correlation benefits could enhance overall performance.

Despite these encouraging results, several limitations warrant consideration. First, while our analysis accounts for transaction costs, implementation challenges such as market impact and capacity constraints in real-world trading scenarios require further investigation. Second, the temporal stability of the EBP signal across different market regimes and economic cycles deserves deeper examination to assess its reliability during periods of market stress or structural changes. Third, our analysis focuses primarily on U.S. equity markets, leaving the international applicability of EBP as an open question.

Future research should explore several promising avenues. Investigating the economic mechanisms underlying the EBP premium would enhance our theoretical understanding of why this signal predicts returns. Does it capture a risk premium associated with financial distress or capital structure adjustments, or does it reflect behavioral biases in how investors process balance sheet information? Additionally, examining the interaction between EBP and corporate events such as earnings announcements, debt issuances, or dividend policies could provide insights into the signal's information content. International studies would help establish whether EBP's predictive power extends beyond U.S. markets, while analysis of the signal's performance across different firm characteristics (size, industry, growth stage) could identify where it adds the most value. Finally, exploring optimal portfolio construction techniques that combine EBP with other factors could yield practical insights for multi-factor investing strategies.

In conclusion, Equity Buffer Preservation emerges as a robust addition to the constellation of return-predictive signals, offering both statistical significance and economic value after comprehensive robustness tests. While further research is needed to fully understand its economic foundations and practical boundaries, the evidence presented here strongly supports EBP's inclusion in the toolkit of cross-sectional return prediction strategies.

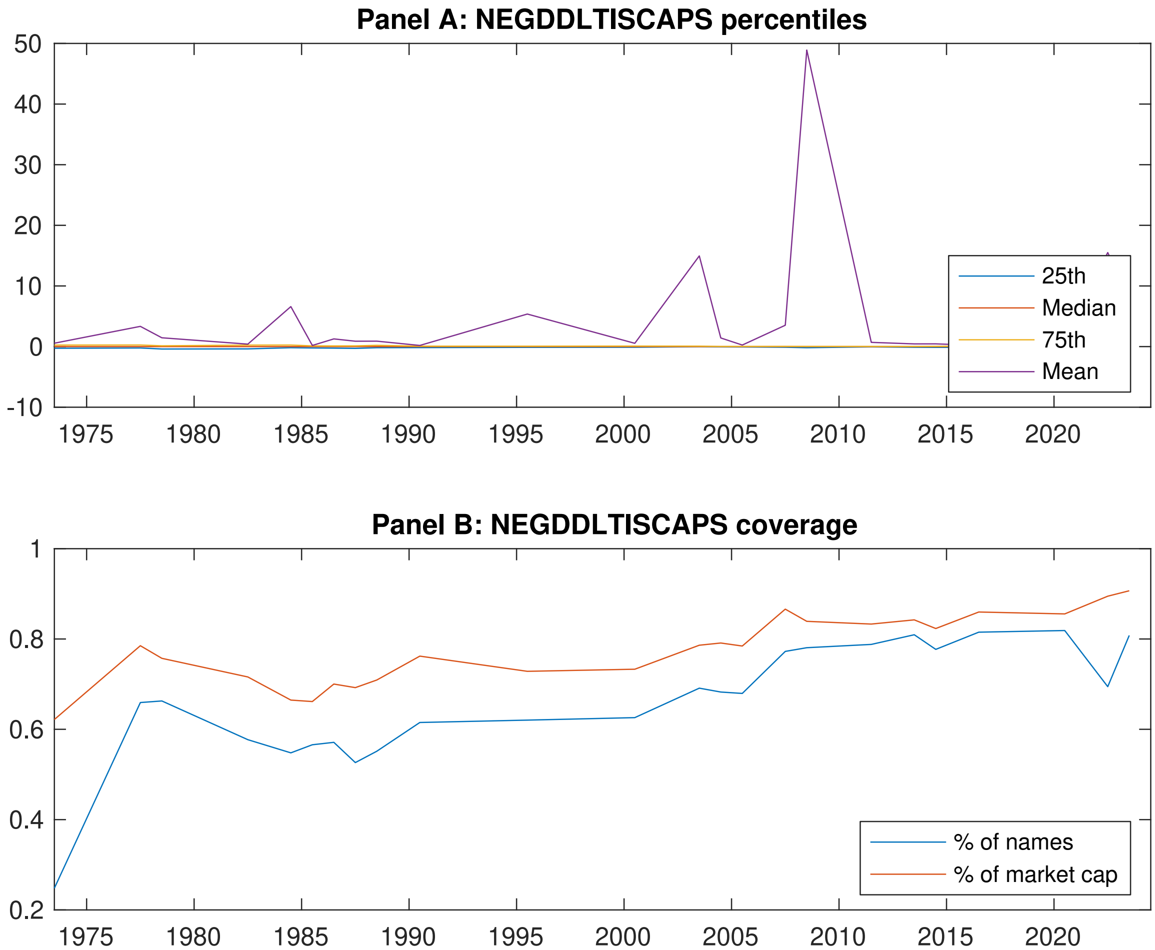


Figure 1: Times series of EBP percentiles and coverage.
This figure plots descriptive statistics for EBP. Panel A shows cross-sectional percentiles of EBP over the sample. Panel B plots the monthly coverage of EBP relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EBP. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on EBP-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.59	0.65	0.61	0.68	0.86	0.28
	[2.99]	[3.33]	[2.93]	[3.67]	[4.55]	[3.77]
α_{CAPM}	-0.07	-0.02	-0.08	0.05	0.22	0.29
	[-1.30]	[-0.32]	[-1.17]	[1.05]	[4.27]	[3.98]
α_{FF3}	-0.11	-0.04	-0.01	0.03	0.19	0.31
	[-2.09]	[-0.75]	[-0.08]	[0.74]	[3.75]	[4.17]
α_{FF4}	-0.09	-0.01	0.06	0.03	0.15	0.24
	[-1.61]	[-0.26]	[0.99]	[0.59]	[2.93]	[3.26]
α_{FF5}	-0.18	-0.02	0.12	-0.00	0.09	0.27
	[-3.22]	[-0.40]	[1.88]	[-0.09]	[1.81]	[3.58]
α_{FF6}	-0.15	-0.00	0.16	-0.00	0.07	0.22
	[-2.78]	[-0.08]	[2.62]	[-0.11]	[1.33]	[2.96]
Panel B: Fama and French (2018) 6-factor model loadings for EBP-sorted portfolios						
β_{MKT}	1.04	1.02	0.98	0.98	1.03	-0.01
	[81.98]	[91.48]	[67.48]	[90.81]	[89.02]	[-0.73]
β_{SMB}	-0.01	-0.06	-0.01	0.00	0.06	0.07
	[-0.27]	[-3.56]	[-0.31]	[0.05]	[3.52]	[2.58]
β_{HML}	0.10	0.11	-0.16	-0.01	-0.02	-0.12
	[4.02]	[5.12]	[-5.69]	[-0.36]	[-1.09]	[-3.72]
β_{RMW}	0.18	0.03	-0.19	0.03	0.12	-0.06
	[7.12]	[1.32]	[-6.56]	[1.53]	[5.17]	[-1.80]
β_{CMA}	0.01	-0.11	-0.19	0.11	0.23	0.23
	[0.15]	[-3.25]	[-4.38]	[3.47]	[6.85]	[4.52]
β_{UMD}	-0.04	-0.02	-0.07	0.00	0.04	0.07
	[-2.89]	[-2.17]	[-4.99]	[0.13]	[3.30]	[4.38]
Panel C: Average number of firms (n) and market capitalization (me)						
n	486	560	1031	619	471	
me ($\$10^6$)	1724	2516	1899	2445	1689	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the EBP strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.28	0.29	0.31	0.24	0.27	0.22
			[3.77]	[3.98]	[4.17]	[3.26]	[3.58]	[2.96]
Quintile	NYSE	EW	0.22	0.25	0.23	0.22	0.19	0.19
			[4.96]	[5.50]	[5.22]	[4.75]	[4.38]	[4.17]
Quintile	Name	VW	0.24	0.27	0.27	0.21	0.21	0.17
			[3.53]	[4.00]	[4.04]	[3.16]	[3.09]	[2.52]
Quintile	Cap	VW	0.23	0.26	0.27	0.22	0.24	0.20
			[3.25]	[3.63]	[3.85]	[3.03]	[3.28]	[2.72]
Decile	NYSE	VW	0.24	0.28	0.27	0.23	0.20	0.18
			[2.82]	[3.32]	[3.19]	[2.76]	[2.31]	[2.08]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.23	0.23	0.24	0.21	0.21	0.18
			[3.09]	[3.16]	[3.31]	[2.84]	[2.82]	[2.44]
Quintile	NYSE	EW	0.01	0.03	0.01	0.01		
			[0.13]	[0.56]	[0.25]	[0.17]		
Quintile	Name	VW	0.19	0.22	0.22	0.19	0.17	0.14
			[2.81]	[3.26]	[3.28]	[2.83]	[2.49]	[2.14]
Quintile	Cap	VW	0.19	0.20	0.22	0.18	0.18	0.15
			[2.59]	[2.84]	[3.02]	[2.57]	[2.53]	[2.18]
Decile	NYSE	VW	0.19	0.23	0.22	0.20	0.16	0.14
			[2.22]	[2.73]	[2.64]	[2.42]	[1.88]	[1.71]

Table 3: Conditional sort on size and EBP

This table presents results for conditional double sorts on size and EBP. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EBP. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EBP and short stocks with low EBP .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results

	EBP Quintiles					EBP Strategies					
Size quintiles	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
(1)	0.75	0.74	0.92	0.79	0.88	0.13	0.14	0.13	0.07	0.08	0.05
	[2.89]	[2.53]	[3.35]	[2.64]	[3.23]	[1.32]	[1.50]	[1.33]	[0.73]	[0.86]	[0.47]
(2)	0.78	0.85	0.88	0.85	0.86	0.08	0.11	0.09	0.07	0.06	0.05
	[3.02]	[3.21]	[3.36]	[3.31]	[3.47]	[1.09]	[1.54]	[1.18]	[0.97]	[0.77]	[0.65]
(3)	0.76	0.82	0.81	0.77	0.92	0.16	0.19	0.18	0.14	0.16	0.13
	[3.14]	[3.46]	[3.26]	[3.28]	[3.93]	[2.07]	[2.54]	[2.41]	[1.85]	[2.05]	[1.67]
(4)	0.69	0.90	0.81	0.77	0.82	0.12	0.14	0.13	0.10	0.09	0.07
	[3.17]	[4.09]	[3.58]	[3.54]	[3.82]	[1.66]	[1.87]	[1.67]	[1.35]	[1.11]	[0.93]
(5)	0.54	0.55	0.57	0.63	0.83	0.29	0.31	0.32	0.25	0.27	0.21
	[2.81]	[2.82]	[2.86]	[3.27]	[4.47]	[3.28]	[3.45]	[3.66]	[2.83]	[2.95]	[2.39]

Panel B: Portfolio average number of firms and market capitalization

	EBP Quintiles: Average n					EBP Quintiles: Average market capitalization ($\$10^6$)				
Size quintiles	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
(1)	358	356	358	357	355	36	29	30	29	34
(2)	98	98	97	98	98	56	56	55	56	56
(3)	69	69	69	69	69	99	100	96	97	99
(4)	58	58	58	58	58	215	218	212	214	212
(5)	53	52	53	52	53	1485	1882	1513	1898	1498

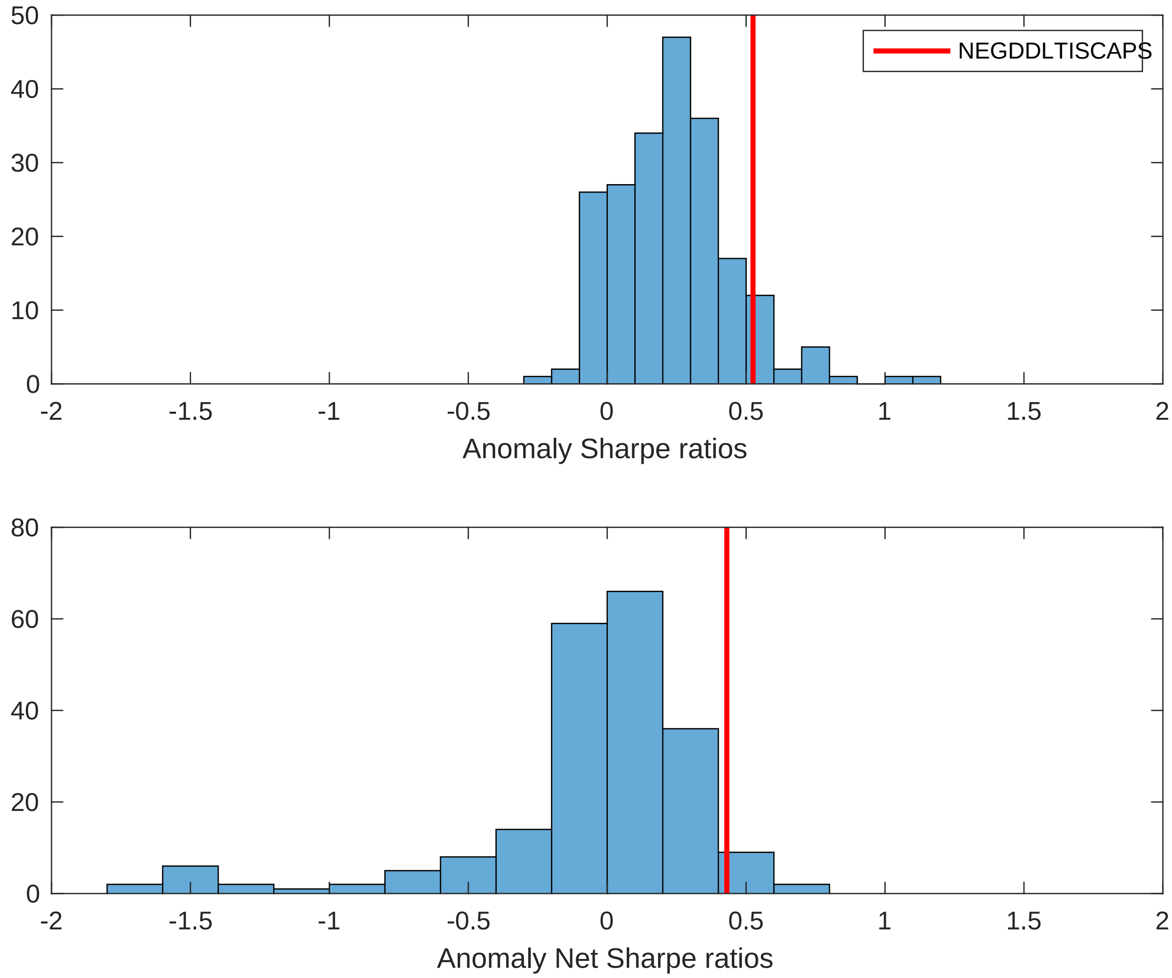


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EBP with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

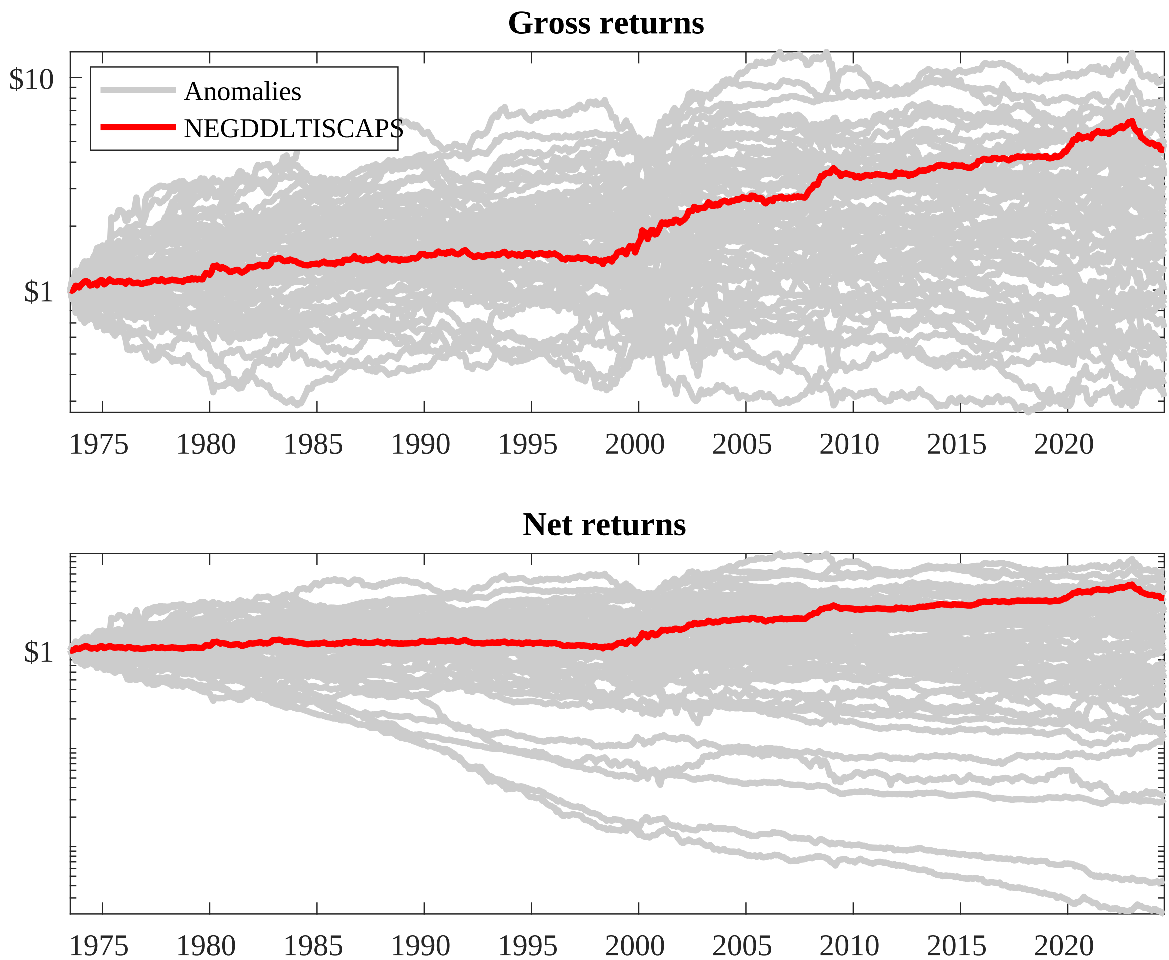
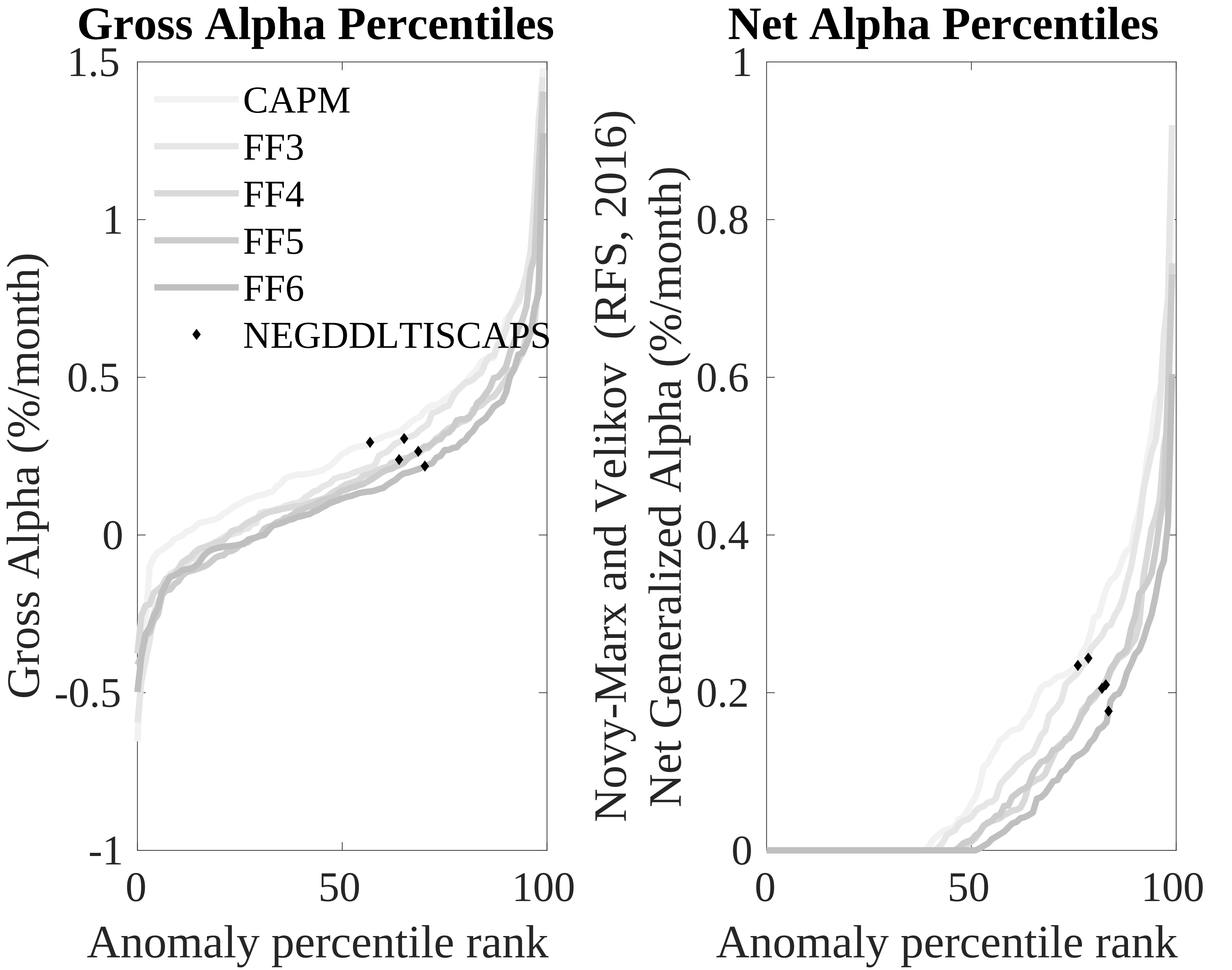


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EBP trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.



Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
NEGDDLTISCAPS
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
0
50
100
1
0.8
0.6
0.4
0.2
0
0
50
100

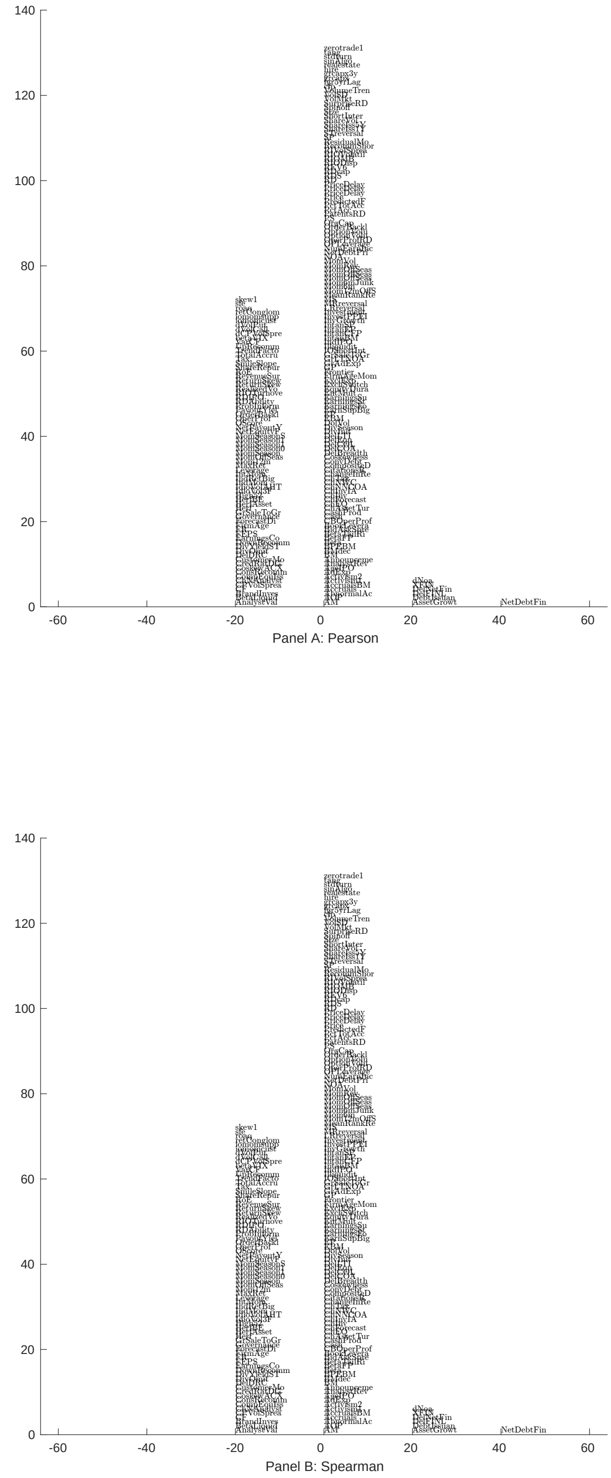


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with EBP. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

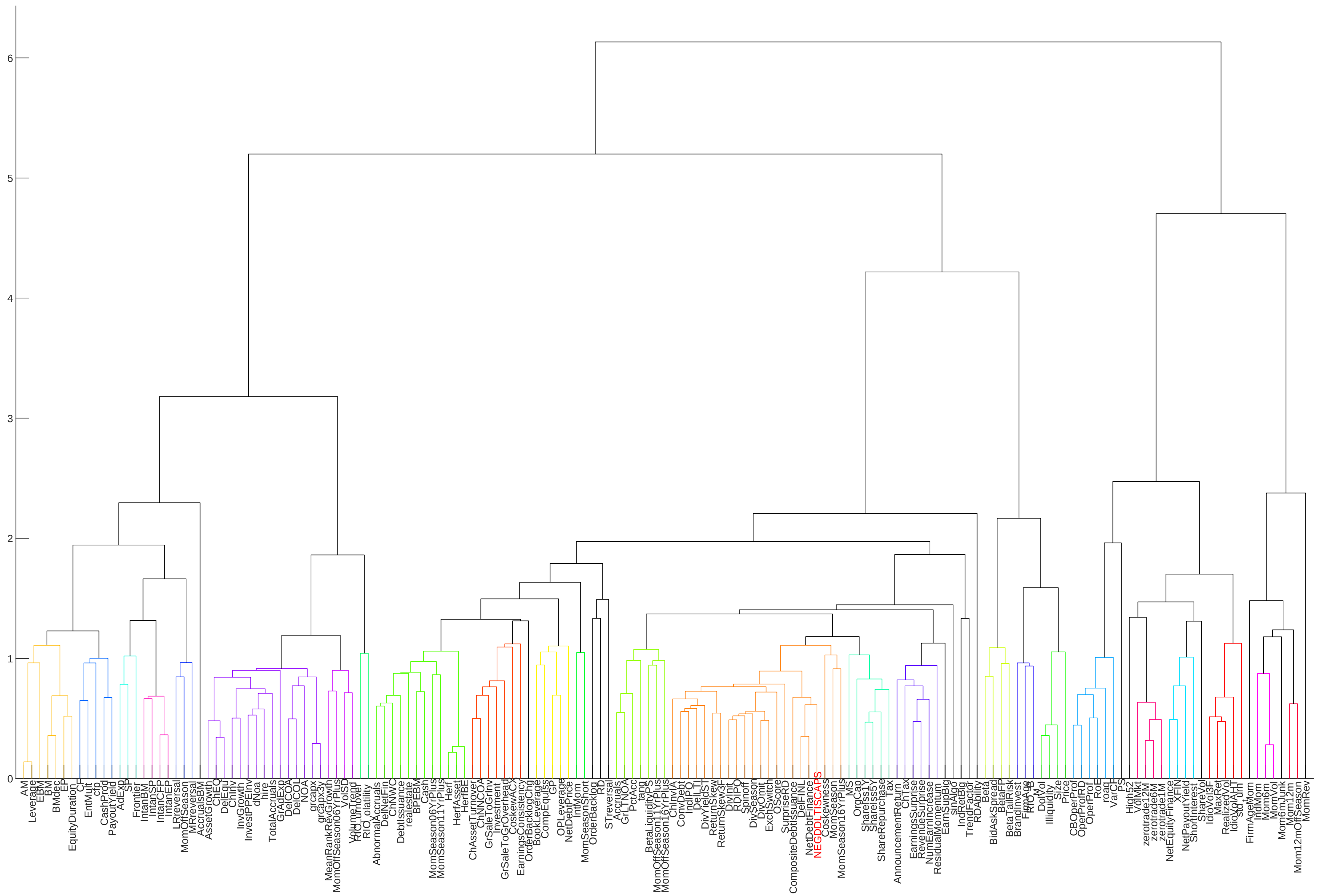


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

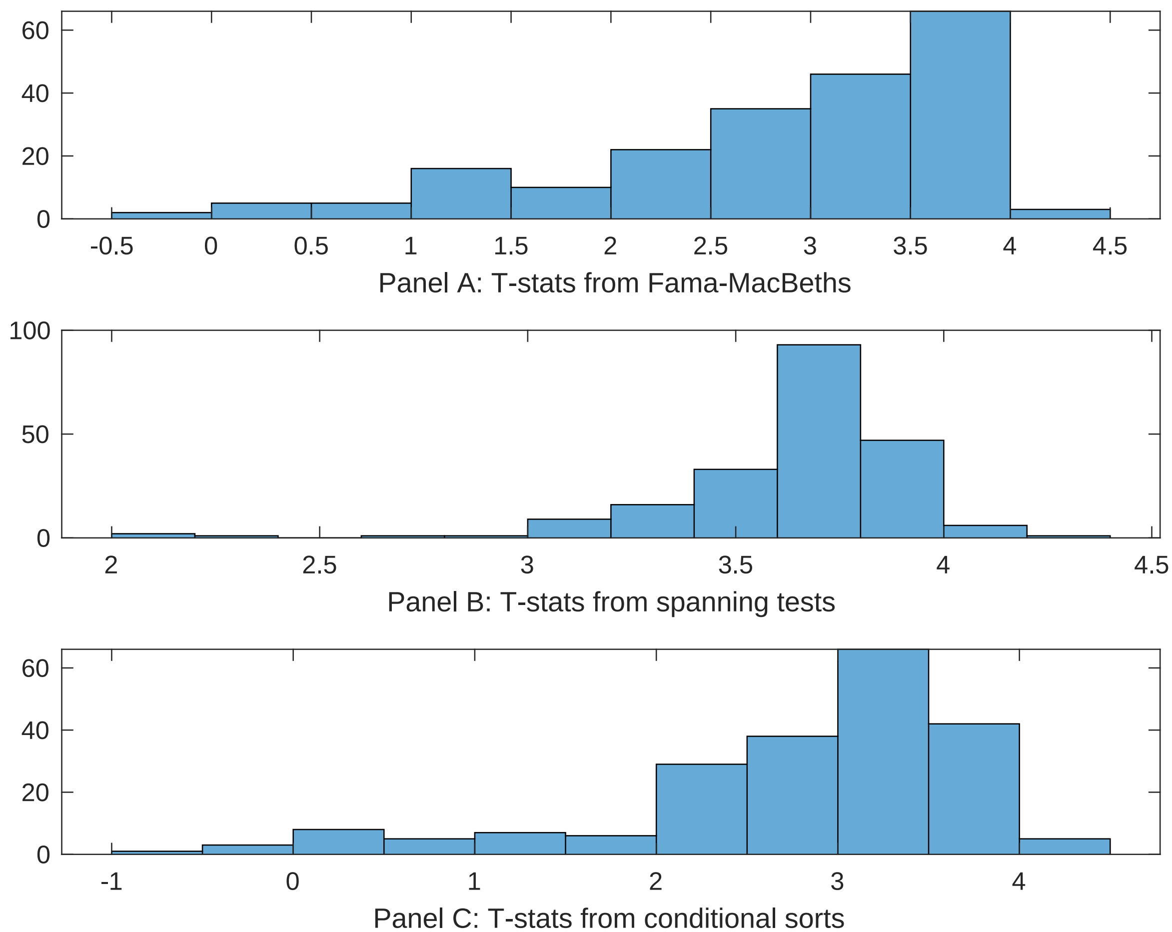


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of EBP conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EBP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EBP} EBP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EBP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EBP. Stocks are finally grouped into five EBP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EBP trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies
This table presents Fama-MacBeth results of returns on EBP. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EBP}EBP_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k}X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Change in financial liabilities, Net debt financing, Inventory Growth, Investment to revenue, Accruals, Book leverage (annual). These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13	0.13	0.13	0.16	0.12	0.13	0.16
	[5.28]	[5.25]	[5.22]	[6.27]	[4.92]	[5.19]	[6.47]
EBP	0.30	0.24	0.16	0.15	0.18	0.19	0.28
	[0.53]	[0.43]	[2.48]	[3.11]	[3.42]	[3.69]	[0.40]
Anomaly 1	0.19						-0.18
	[9.45]						[-0.04]
Anomaly 2		0.21					0.14
		[8.96]					[2.73]
Anomaly 3			0.43				0.27
			[7.25]				[4.08]
Anomaly 4				0.26			0.24
				[5.99]			[4.60]
Anomaly 5					0.14		0.74
					[4.38]		[1.80]
Anomaly 6						0.12	0.45
						[1.27]	[0.38]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies
This table presents spanning tests results of regressing returns to the EBP trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EBP} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Change in financial liabilities, Net debt financing, Inventory Growth, Investment to revenue, Accruals, Book leverage (annual). These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.19	0.19	0.20	0.19	0.19	0.18	0.17
	[2.58]	[2.62]	[2.80]	[2.67]	[2.59]	[2.48]	[2.38]
Anomaly 1	12.32						-2.36
	[2.98]						[-0.42]
Anomaly 2		14.59					16.87
		[3.67]					[3.14]
Anomaly 3			4.82				4.59
			[1.69]				[1.64]
Anomaly 4				6.12			4.97
				[2.15]			[1.76]
Anomaly 5					3.85		-3.84
					[1.32]		[-1.17]
Anomaly 6						8.99	11.61
						[3.41]	[3.79]
mkt	-0.68	-0.94	-1.06	-1.10	-0.68	0.15	-0.01
	[-0.41]	[-0.56]	[-0.63]	[-0.65]	[-0.40]	[0.09]	[-0.00]
smb	5.64	5.62	7.43	5.77	7.64	5.96	3.28
	[2.19]	[2.20]	[2.90]	[2.22]	[2.93]	[2.34]	[1.19]
hml	-11.15	-11.38	-12.12	-11.55	-10.98	-5.37	-3.75
	[-3.44]	[-3.53]	[-3.73]	[-3.55]	[-3.30]	[-1.43]	[-1.00]
rmw	-6.86	-7.36	-5.06	-5.24	-4.49	-3.36	-4.47
	[-2.08]	[-2.23]	[-1.52]	[-1.59]	[-1.31]	[-1.00]	[-1.29]
cma	18.92	19.26	18.75	22.46	21.79	22.96	16.08
	[3.71]	[3.85]	[3.38]	[4.55]	[4.34]	[4.69]	[2.86]
umd	6.53	6.65	7.10	6.81	7.31	7.41	5.64
	[3.83]	[3.96]	[4.18]	[3.98]	[4.32]	[4.45]	[3.26]
# months	612	612	612	612	612	612	612
$\bar{R}^2$(%)	11	12	10	11	10	12	14

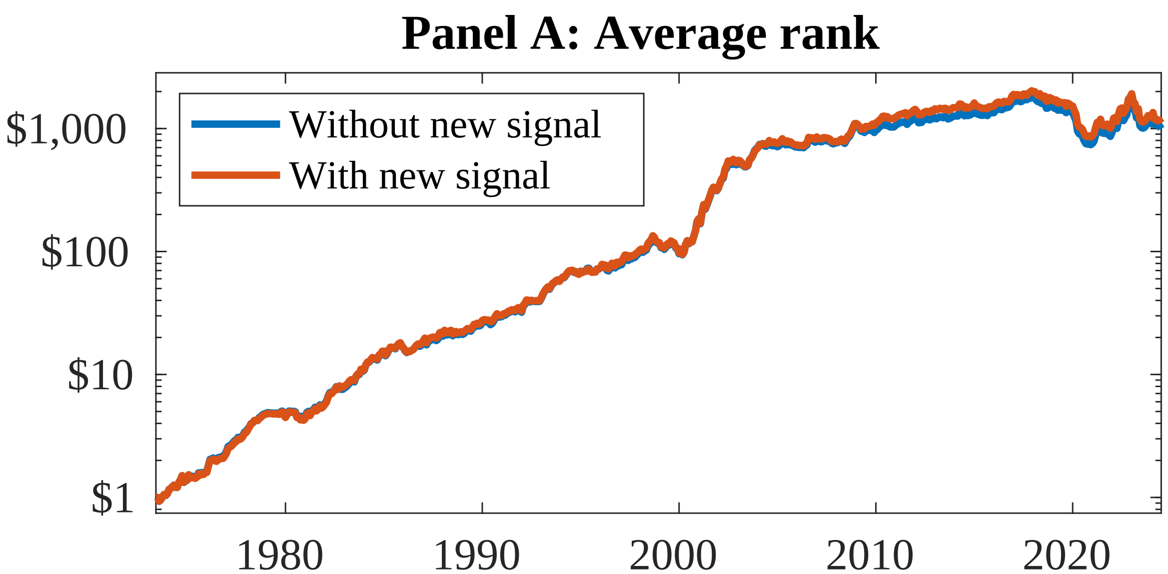


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as EBP. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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